Mathematical introduction

Framework, Markov chains, and methods

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Chapter [Mathematical introduction]

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Contents

Chapter 2

Mathematical introduction

2.1 Overview

This chapter sets out the mathematical framework used throughout the thesis: Markovian population models, discrete or continuous in time, in which counts may be small enough that extinction, establishment, and absorption are random events rather than the deterministic outcomes of a rate equation. When the mean trajectory answers the wrong question, one needs a distribution, and the apparatus below is chosen for that purpose.

The chapter is organised as objects first, methods second.

Section 2.2 surveys the process classes that appear later. In discrete time the main examples are simple random walks and Galton–Watson branching (the latter only briefly here). In continuous time the building blocks are exponential clocks and the chains assembled from them: Poisson processes, birth–death processes, and birth–death–catastrophe processes. Time-inhomogeneous models and systems that couple ordinary differential equations to continuous-time Markov chains complete the inventory. Notation for generators, competing clocks, and probability generating functions is fixed along the way.

Section 2.3 collects the analytic methods applied to those processes. Discrete-time devices—first-step analysis and functional iteration of generating functions—are treated briefly; a fuller development of the quasi-stationary amplitude $A(p)$ and of Koenigs-function methods is deferred to Chapter [The constant $A(p)$]. Continuous-time methods are developed with linear birth–death processes as the running example: backward equations, means and higher moments, hitting and extinction probabilities, and conditional means, including the continuous-time constant $A_{\mathrm{c}}(p)$ and a light definition of the discrete constant $A(p)$. The method of characteristics is presented with one worked example; extended absorption models are recorded in the appendix.

Two modelling threads run through later chapters and are introduced only as far as the toolkit requires. The first is *conditioning on survival*: a subcritical branching process dies out almost surely, yet conditioned on non-extinction its population settles to a quasi-stationary law whose mean is $1/A(p)$. The second is the *near-critical regime*, in which lifetimes become long and numerical schemes that iterate to stationarity become expensive. Deep analysis of $A(p)$ itself—product and series representations, bounds, closed-form search, and the Koenigs obstruction—belongs to Chapter [The constant $A(p)$], not to the present chapter.

Open questions that sit with this toolkit rather than with the arithmetic of $A(p)$ include the existence of a quasi-stationary distribution when one conditions on neither extinction nor rupture, and the tightness of Jensen-type bounds for discrete catastrophe survival. Those are left for later modelling chapters.

2.2 Markov chains

This section names the process classes used in the thesis and fixes their basic structure. Methods of analysis are deferred to section 2.3.

2.2.1 Discrete-time Markov chains

A discrete-time Markov chain on a countable state space $\mathcal{S}$ is a process $(X_n)_{n\geq 0}$ satisfying

$$\mathbb{P}(X_{n+1} = j \mid X_n = i,\, X_{n-1}, \dots, X_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i) = P_{ij}.$$

The matrix $P = (P_{ij})$ is the one-step transition kernel. Absorbing states, hitting probabilities, and first-step equations are developed as methods in section 2.3.1; the two examples below supply the chains to which those methods are applied.

2.2.1.1 Simple random walks

Remark 2.1 (Scope). What follows is the standard definition for later use; hitting probabilities are treated as a method in section 2.3.2.

Let $(S_n)_{n\geq 0}$ be a simple symmetric random walk on $\mathbb{Z}$, $S_{n+1} = S_n + \xi_{n+1}$ with $\mathbb{P}(\xi = \pm 1) = \frac{1}{2}$, started at $S_0 = x$. The one-step transitions are then $P_{x,x+1} = P_{x,x-1} = \frac{1}{2}$. On a finite interval with absorbing endpoints (gambler's ruin), first-step analysis (section 2.3.1) yields the classical linear hitting probabilities. Biased walks and continuous-time analogues reappear as embedded jump chains of birth–death processes below.

Figure 2.1 records the standard absorbing chain on $\{0, 1, \dots, N\}$ together with the hitting probabilities $h_i = \mathbb{P}_i(\tau_N < \tau_0)$ of reaching the goal N before ruin at 0. For the symmetric walk one has the linear law $h_i = i/N$; with upward step probability $p \neq \frac{1}{2}$ and $r = q/p$, $q = 1 - p$, the same first-step system gives the curved profile $h_i = (r^i - 1)/(r^N - 1)$.

2.2.1.2 Galton–Watson processes

A Galton–Watson process $(Z_n)_{n\geq 0}$ is a discrete-time Markov chain on $\{0, 1, 2, \dots\}$ in which each individual independently produces offspring according to a fixed law, and the next generation is the sum of those contributions. With offspring probability generating function $\phi(z) = \mathbb{E}\left(z^L\right)$, the law of Z_n is obtained by iterating ϕ, and the mean satisfies $\mathbb{E}(Z_n) = m^n \mathbb{E}(Z_0)$ when $m = \phi'(1)$.

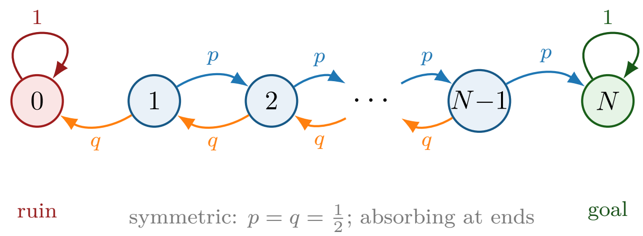
This thesis uses principally the *binary* (death-or-division) law

$$\phi(z) = pz^2 + (1 - p), \qquad p \in [0, 1], \tag{2.1}$$

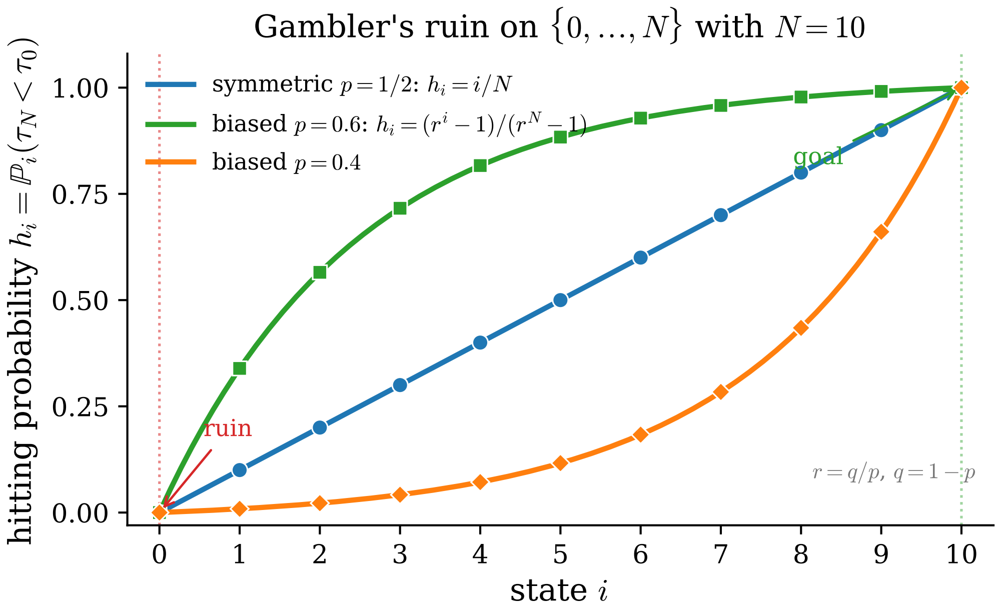
for which $m = 2p$ and the survival probability $S_n = \mathbb{P}(Z_n > 0)$ obeys

$$S_{n+1} = 2p\, S_n - pS_n^2, \qquad S_0 = 1. \tag{2.2}$$

Extinction is certain for $p \leq \frac{1}{2}$. At criticality $p = \frac{1}{2}$ one has the elementary asymptotics $S_n \sim 2/n$ and infinite expected lifetime; in the subcritical regime $p < \frac{1}{2}$ the late-time survival scale involves a constant $A(p)$, introduced lightly in section 2.3.2 and developed in full in Chapter [The constant $A(p)$]. Figure 2.2 summarises the one-step binary law.



(a) Absorbing random walk (gambler's ruin) on $\{0, 1, \ldots, N\}$.



(b) Hitting probabilities $h_i = \mathbb{P}_i(\tau_N < \tau_0)$ for $N = 10$.

Figure 2.1: Gambler's ruin. Panel (a): one-step transitions with absorbing endpoints 0 (ruin) and N (goal). Panel (b): probability of hitting N before 0 from state i, for the symmetric walk $h_i = i/N$ and for biased walks with $p \in \{0.4, 0.6\}$ (where $r = q/p$ and $h_i = (r^i - 1)/(r^N - 1)$). Markers sit at integer states; solid curves are the closed-form profiles.

Remark 2.2 (Greater detail later). Critical lifetime theory, the total-progeny (Catalan) law, phase portraits, and the full quasi-stationary moment calculation for the binary process are deferred: amplitude theory to Chapter [The constant $A(p)$], and modelling-scale applications to later chapters. The present section keeps only the process class and the one-step structure needed for section 2.3.

2.2.2 Continuous-time Markov chains

2.2.2.1 Exponential clocks

Every continuous-time model in this thesis is built from events whose rate of occurrence does not depend on how long the system has already been waiting. That single assumption forces the exponential distribution.

Definition 2.3 (Exponential law)**.** A random variable T is exponentially distributed with rate

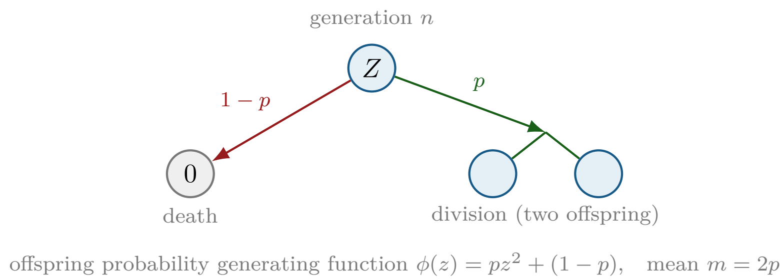


Figure 2.2: Binary Galton–Watson step: each individual independently dies (probability $1-p$) or divides into two (probability p). The next generation is the sum of independent one-step contributions.

$\theta > 0$, written $T \sim \mathrm{Exp}(\theta)$, if it has density and distribution function

$$f_T(t) = \theta \mathrm{e}^{-\theta t}, \qquad F_T(t) = 1 - \mathrm{e}^{-\theta t}, \qquad t \geq 0,$$

and $f_T(t) = 0$ for $t < 0$. Its mean is $1/\theta$ and its variance $1/\theta^2$.

The property that matters is memorylessness. For $t_1, t_2 > 0$,

$$\mathbb{P}(T > t_1 + t_2 \mid T > t_1) = \frac{\mathrm{e}^{-\theta(t_1+t_2)}}{\mathrm{e}^{-\theta t_1}} = \mathrm{e}^{-\theta t_2} = \mathbb{P}(T > t_2)\,, \tag{2.3}$$

so a clock that has already run for time t_1 is statistically indistinguishable from a fresh one. The exponential law is the only continuous distribution with this property, which is why a process assembled from constant-rate events depends on its present state alone and not on its history.

When several events compete, the following two facts organise everything that follows. Let $T_1, \ldots, T_m$ be independent with $T_i \sim \mathrm{Exp}(\theta_i)$, and let $T = \min_i T_i$ be the time of the first event.

Proposition 2.4 (Competing clocks)**.** *With the notation above, write $\Theta = \sum_{i=1}^m \theta_i$. Then*

$$T \sim \mathrm{Exp}(\Theta), \qquad \mathbb{P}(T_k = T) = \frac{\theta_k}{\Theta}, \tag{2.4}$$

and the identity of the winning clock is independent of T.

Proof. For the first claim, $\mathbb{P}(T > t) = \prod_i \mathbb{P}(T_i > t) = \prod_i \mathrm{e}^{-\theta_i t} = \mathrm{e}^{-\Theta t}$. For the second, condition on the value of T_k and use independence:

$$\mathbb{P}(T_k = T) = \int_0^\infty \theta_k \mathrm{e}^{-\theta_k u} \prod_{i \neq k} \mathrm{e}^{-\theta_i u}\, \mathrm{d}u = \theta_k \int_0^\infty \mathrm{e}^{-\Theta u}\, \mathrm{d}u = \frac{\theta_k}{\Theta}.$$

Repeating the calculation with the integral truncated at t gives $\mathbb{P}(T_k = T,\ T > t) = (\theta_k/\Theta)\mathrm{e}^{-\Theta t} = \mathbb{P}(T_k = T)\,\mathbb{P}(T > t)$, which is the asserted independence. □

Proposition 2.4 is used in two ways. First, a rate θ_k becomes a probability θ_k/Θ as soon as one asks which of several things happened rather than when; section 2.3.2 uses exactly this to identify the replication probability p of a discrete-generation model with $\lambda/(\lambda+\mu)$ in a continuous-time one. Second, the same proposition is the entire content of the direct simulation method.

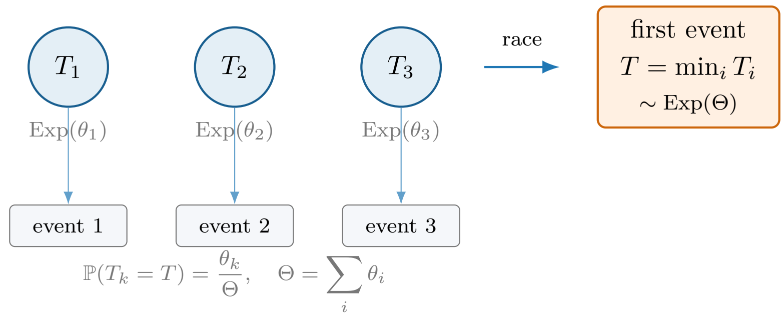


Figure 2.3: Competing exponential clocks (proposition 2.4): independent holding times race; the minimum is exponential with rate Θ, and the winning event is chosen with probability proportional to its rate, independently of the waiting time.

Remark 2.5 (Direct simulation). Proposition 2.4 is an algorithm as well as a theorem. Given the current state, compute the rate θ_i of every possible event and their sum Θ; draw $T \sim \mathrm{Exp}(\Theta)$ by inverting the distribution function of definition 2.3 on a uniform variate; draw the event k with probability θ_k/Θ; advance the clock by T, apply event k, and repeat. Because the two draws are independent and the exponential law is memoryless, the resulting trajectory has exactly the law of the process. This is Gillespie's direct method [1]; in this thesis it is used to check analytical results, never to replace them.

2.2.2.2 Generators and master equations

Let $\{X_t : t \geq 0\}$ be a stochastic process taking values in a countable state space $\mathcal{S}$, which throughout will be a set of population counts or of tuples of counts. The process is a *continuous-time Markov chain* if, for all times $s, t \geq 0$ and all states,

$$\mathbb{P}(X_{t+s} = j \mid X_t = i,\ \{X_u\}_{u<t}) = \mathbb{P}(X_{t+s} = j \mid X_t = i)\,, \tag{2.5}$$

so that the past influences the future only through the present. The chain is *time homogeneous* if the right-hand side depends on s but not on t, in which case one writes $p_{ij}(s) = \mathbb{P}(X_{t+s} = j \mid X_t = i)$. All chains in this thesis are time homogeneous unless stated otherwise; the time-inhomogeneous case is outlined in section 2.2.3.

Such a chain is specified by its *generator*, the matrix $Q = (q_{ij})$ whose off-diagonal entries are the rates of the corresponding transitions,

$$q_{ij} = \lim_{\Delta t \to 0} \frac{p_{ij}(\Delta t)}{\Delta t} \quad (j \neq i), \qquad q_{ii} = -\sum_{j \neq i} q_{ij} =: -q_i, \tag{2.6}$$

so that every row of Q sums to zero. Equivalently, and this is the form used to define models below,

$$\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i) = q_{ij}\,\Delta t + o(\Delta t) \qquad (j \neq i), \tag{2.7}$$

with the complementary probability $1 - q_i \Delta t + o(\Delta t)$ of no transition at all. By proposition 2.4 the chain leaves state i after a holding time distributed as $\mathrm{Exp}(q_i)$ and then jumps to j with probability q_{ij}/q_i. A state with $q_i = 0$ is *absorbing*: the chain that reaches it stays there forever. Extinction and rupture will both be modelled as absorbing states, and the whole of section 2.3.2 concerns the behaviour of a chain conditioned on not yet having been absorbed.

Differentiating the Chapman–Kolmogorov identity $p_{ij}(t+\Delta t) = \sum_k p_{ik}(t) p_{kj}(\Delta t)$ with respect to Δt at $\Delta t = 0$ gives the *forward Kolmogorov equations*,

$$\frac{\mathrm{d}p_{ij}(t)}{\mathrm{d}t} = \sum_k p_{ik}(t)\, q_{kj}, \qquad \text{that is} \qquad \frac{\mathrm{d}P}{\mathrm{d}t} = P\,Q. \tag{2.8}$$

Written out for a fixed initial state and with $p_j(t)$ for the probability of occupying state j at time t, (2.8) becomes the master equation

$$\frac{\mathrm{d}p_j(t)}{\mathrm{d}t} = \sum_{k\neq j} q_{kj}\, p_k(t) - q_j\, p_j(t) : \tag{2.9}$$

probability flows into state j from every state that can reach it and out of j at total rate q_j. Every model in this thesis is specified by writing down (2.7) and then reading off (2.9).

2.2.2.3 Probability generating functions

For a random variable X taking values in $\{0, 1, 2, \dots\}$ with $p_k = \mathbb{P}(X = k)$, the probability generating function is

$$\phi(z) = \mathbb{E}\Big(z^X\Big) = \sum_{k=0}^{\infty} p_k z^k, \tag{2.10}$$

which converges at least for $|z| \leq 1$ since the p_k sum to one. The series determines the distribution — $p_k = \phi^{(k)}(0)/k!$ — and moments are read off at $z = 1$:

$$\phi(1) = 1, \qquad \phi'(1) = \mathbb{E}(X)\,, \qquad \phi''(1) = \mathbb{E}(X(X-1))\,. \tag{2.11}$$

Two structural properties do most of the work. If X and Y are independent then $\phi_{X+Y} = \phi_X \phi_Y$, so sums become products. And if N is an independent count with probability generating function ψ, the random sum $X_1 + \cdots + X_N$ of independent copies of X has probability generating function $\psi \circ \phi$; composition of generating functions is what makes branching processes tractable, and section 2.2.1.2 rests on it.

For a pair of counts the definition extends to

$$G(x, y) = \mathbb{E}\Big(x^X y^Y\Big) = \sum_i \sum_j \mathbb{P}(X = i,\, Y = j)\; x^i y^j, \tag{2.12}$$

with marginals recovered by setting the other variable to one. The absorption models of section 2.3.3 track particles inside and outside a compartment and therefore need this bivariate form; section 2.C records how to invert it.

When the counts evolve in time, G acquires a time argument, $G(x, y, t) = \mathbb{E}\Big(x^{X_t} y^{Y_t}\Big)$, and the master equation (2.9) for the infinite family of state probabilities collapses to a single partial differential equation for G. The mechanism is uniform: multiply the equation for $p_{ij}(t)$ by $x^i y^j$, sum over all states, shift indices so that every sum is again of the form (2.12), and recognise the factors of i and j produced by the linear rates as derivatives ∂_x and ∂_y. Because the rates in these models are linear in the counts, the resulting equation is always of first order, and the method of characteristics therefore always applies in principle. Section 2.2.2.5 carries this out in the simplest case; section 2.3.3 does it three times over.

2.2.2.4 Poisson processes

Remark 2.6 (Scope). The paragraph below records the standard link between Poisson processes, exponential clocks, and pure birth.

A homogeneous Poisson process $(N_t)_{t\geq 0}$ of rate $\theta > 0$ is the counting process with independent increments and $N_t \sim \text{Poisson}(\theta t)$. Equivalently, the inter-event times are i.i.d. $\text{Exp}(\theta)$. A pure-birth process with constant per-capita birth rate is a Yule process; with a single individual and no death it reduces to a Poisson process of events. Competing exponential clocks (proposition 2.4) build general continuous-time Markov chains from the same holding-time law. Figure 2.4 shows one sample counting path together with the exponential interarrival density that generates it; the mean line $\mathbb{E}(N_t) = \theta t$ provides the deterministic scale against which the jumps fluctuate.

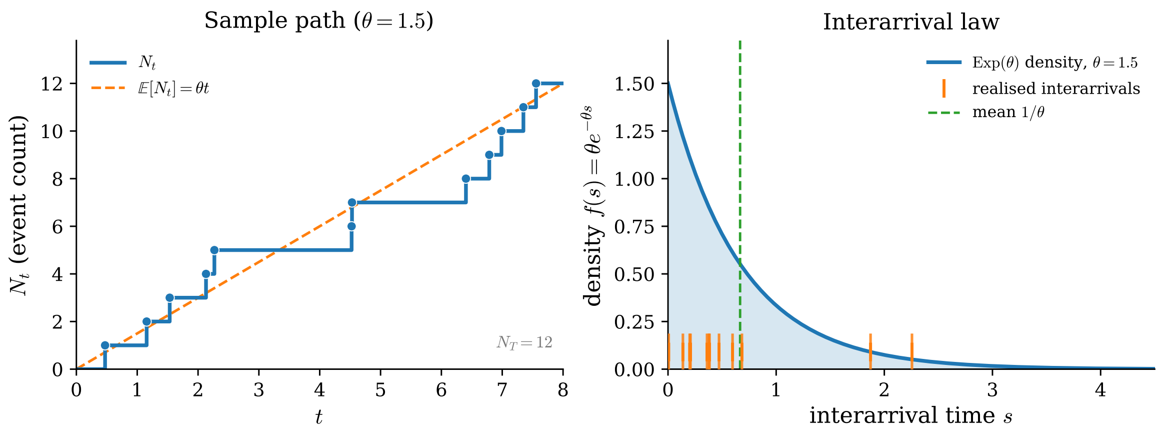


Figure 2.4: Homogeneous Poisson process of rate $\theta = 1.5$. Left: a sample path of the counting process N_t (step function) with event markers, and the mean trajectory $\mathbb{E}(N_t) = \theta t$ (dashed). Right: density $f(s) = \theta \mathrm{e}^{-\theta s}$ of the i.i.d. $\text{Exp}(\theta)$ interarrival times, with the realised waits from the left-hand path marked as a rug and the mean wait $1/\theta$ indicated.

2.2.2.5 Birth–death processes

The linear birth–death process is the continuous-time counterpart of the Galton–Watson process, and it is the last piece of standard machinery needed. Let X_t count individuals, each of which independently gives birth at rate λ and dies at rate μ. In the notation of (2.7) the non-zero rates out of state n are

$$q_{n,n+1} = \lambda n, \qquad q_{n,n-1} = \mu n, \tag{2.13}$$

so that $q_n = (\lambda + \mu)n$ and $n = 0$ is absorbing; see the rate diagram in figure 2.5. The master equation (2.9) reads

$$\frac{\mathrm{d}p_n(t)}{\mathrm{d}t} = \lambda(n-1)p_{n-1}(t) + \mu(n+1)p_{n+1}(t) - (\lambda+\mu)n\, p_n(t). \tag{2.14}$$

Multiplying by z^n and summing over $n \geq 0$, the three terms become $\lambda z^2 \partial_z G$, $\mu\, \partial_z G$ and $-(\lambda+\mu)z\, \partial_z G$ after the index shifts described above, so that

$$\frac{\partial G}{\partial t} = (\lambda z^2 - (\lambda+\mu)z + \mu)\frac{\partial G}{\partial z} = (\lambda z - \mu)(z-1)\frac{\partial G}{\partial z}, \qquad G(z,0) = z^N, \tag{2.15}$$

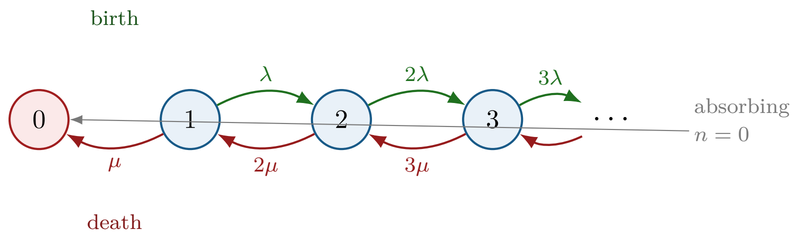


Figure 2.5: Linear birth–death process on $\{0, 1, 2, \dots\}$: per-capita rates λ (birth) and μ (death) yield transition rates $q_{n,n+1} = \lambda n$ and $q_{n,n-1} = \mu n$. State 0 is absorbing (extinction).

for a process started from N individuals. The quadratic coefficient factorises with roots $z = 1$ and $z = \mu/\lambda$; the same factorisation, in a different variable, reappears in section 2.A.2 and is what makes that calculation possible. Solving (2.15) along its characteristics gives the classical result of Kendall [2],

$$G(z,t) = \left(\frac{\mu(1-z) - (\mu - \lambda z)\mathrm{e}^{-(\lambda-\mu)t}}{\lambda(1-z) - (\mu - \lambda z)\mathrm{e}^{-(\lambda-\mu)t}} \right)^N, \qquad \lambda \neq \mu. \tag{2.16}$$

Two consequences are needed later. The mean follows from (2.11) or, more quickly, from the fact that each individual contributes independently:

$$\mathbb{E}(X_t) = N\mathrm{e}^{(\lambda-\mu)t}. \tag{2.17}$$

The probability of extinction by time t is $p_0(t) = G(0,t)$, that is

$$p_0(t) = \begin{cases} \left(\dfrac{\mu - \mu\mathrm{e}^{(\mu-\lambda)t}}{\lambda - \mu\mathrm{e}^{(\mu-\lambda)t}} \right)^N, & \lambda \neq \mu, \\ \left(\dfrac{\lambda t}{1 + \lambda t} \right)^N, & \lambda = \mu, \end{cases} \tag{2.18}$$

the critical case following by taking $\mu \to \lambda$ in the first line. Letting $t \to \infty$ recovers the familiar threshold: extinction is certain when $\lambda \leq \mu$, and occurs with probability $(\mu/\lambda)^N$ when $\lambda > \mu$. Figure 2.6 plots the mean and survival curves in the three regimes; section 2.3.2 takes (2.18) as its starting point.

2.2.2.6 Birth–death–catastrophe processes

The discrete rupture mechanism of earlier drafts has a continuous-time counterpart, and since that process is the central object of several later chapters it is defined here in the notation already fixed. No results about it are derived in this chapter.

Definition 2.7 (Birth–death–catastrophe process)**.** Let X_t count individuals in a compartment. Each individual independently gives birth at rate λ and dies at rate μ, and the compartment additionally undergoes a catastrophe — a rupture that removes the entire population at once — at a rate ρ per individual, so that the total catastrophe rate in state n is ρn. Writing † for the post-catastrophe state, the non-zero transition rates out of $n \geq 1$ are

$$q_{n,n+1} = \lambda n, \qquad q_{n,n-1} = \mu n, \qquad q_{n,\dagger} = \rho n, \tag{2.19}$$

and both $n = 0$ and † are absorbing (figure 2.7).

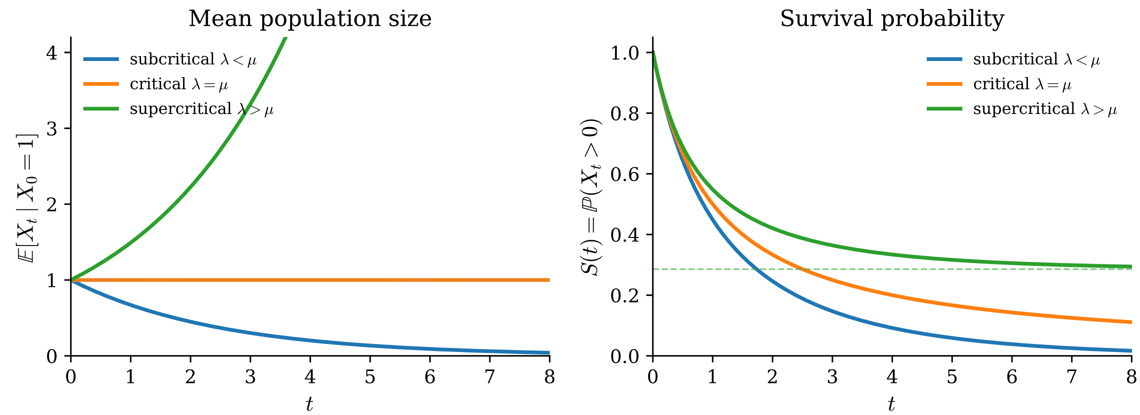


Figure 2.6: Linear birth–death process started from one individual: mean population size (left) and survival probability $S(t) = \mathbb{P}(X_t > 0)$ (right) for subcritical ($\lambda = 0.6$, $\mu = 1$), critical ($\lambda = \mu = 1$), and supercritical ($\lambda = 1.4$, $\mu = 1$) rates. In the supercritical case the survival probability approaches $1 - \mu/\lambda$ (dashed).

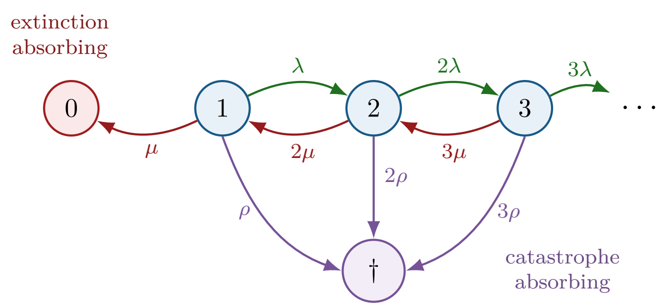


Figure 2.7: Birth–death–catastrophe process: the birth–death chain on the positive integers is killed at per-capita rate ρ, sending the process to a second absorbing state † (catastrophe/rupture). Extinction (0) and catastrophe (†) are distinct absorbing mechanisms.

Two features distinguish definition 2.7 from the linear birth–death process of section 2.2.2.5, and both matter later. The process now has *two* absorbing states, reached by mechanisms that are distinguishable and not interchangeable: extinction empties the compartment gradually, catastrophe empties it in one event and releases its contents. And the catastrophe rate is proportional to the population, so the compartment becomes more likely to rupture the fuller it is.

The generating-function machinery of section 2.2.2.3 applies unchanged. Restricting attention to realisations that have not yet ruptured and writing $G(z,t) = \mathbb{E}\Big(z^{X_t}\mathbf{1}\{X_t \neq \dagger\}\Big)$, the master equation for (2.19) yields

$$\frac{\partial G}{\partial t} = (\lambda z^2 - (\lambda + \mu + \rho)z + \mu)\frac{\partial G}{\partial z}, \qquad G(z,0) = z^N, \tag{2.20}$$

which differs from (2.15) only in the coefficient of z. The quadratic no longer factorises with a root at $z = 1$, and that single change is responsible for most of what is interesting about the process: probability is not conserved on the non-ruptured states, and $G(1,t)$ is the survival probability rather than one. Rates, means and variances, state probabilities and the associated generating-function equations are developed in the later chapters on birth–death–catastrophe processes, with the two-type extension treated in the chapter on multi-type processes and the release of a ruptured compartment's contents into a shared medium in the chapter on compartment rupture. The method used to solve (2.20) and its relatives is the subject of section 2.3.3.

Discrete rupture as a killed chain

The continuous-time process of definition 2.7 may also be viewed as a birth–death process killed at a state-dependent rate. Restricting the generator to the positive states produces a killed subgenerator $Q^{\dagger}$ whose off-diagonal entries are the birth and death rates and whose diagonal absorbs the catastrophe rate ρn. A quasi-stationary distribution ν on $\{1, 2, \dots\}$, when it exists, satisfies the left eigenrelation

$$\nu Q^{\dagger} = -\gamma\, \nu, \qquad \nu \mathbf{1} = 1, \tag{2.21}$$

for some decay parameter $\gamma > 0$: under ν, the residual lifetime to absorption (by death or by catastrophe) is exponential with rate γ, and the conditional law of the population given non-absorption is stationary. The present chapter needs only the distinction between the two absorbing mechanisms and the form of (2.21); rates, means and the full spectral theory belong to later modelling chapters.

Remark 2.8 (Catastrophe is not a mean-field hazard). A catastrophe rate proportional to population size is a genuine jump mechanism on the state space, not a deterministic hazard applied to the mean trajectory. Replacing the random census Z_i by $\mathbb{E}(Z_i)$ in discrete survival formulae such as replacing the random census by its mean systematically understates survival (Jensen), and confuses non-rupture with non-extinction. The two events are distinct, and joint conditioning on neither is a separate problem.

2.2.3 Time-inhomogeneous processes

2.2.3.1 Logistic speciation model

Remark 2.9 (TODO — logistic speciation). No logistic speciation (time-inhomogeneous) model is developed in this chapter yet. Intended content, pending notes or references: a continuous-time process whose birth or death rates depend explicitly on absolute time (or on a deterministic environmental schedule), with logistic density dependence in the speciation or birth rate; the generator form; contrast with the time-homogeneous birth–death process of section 2.2.2.5; and a pointer to any later chapter that uses the model.

2.2.4 Coupled ODE–CTMC systems

Remark 2.10 (TODO — concrete coupling examples). The framework below is abstract; concrete examples are deferred to the modelling chapters that use them.

Several models in the wider thesis couple a continuous-time Markov chain (X_t) on a discrete state space to an ordinary differential equation for a continuous state $y_t \in \mathbb{R}^d$. Three dependence patterns arise.

(i) *ODE driven by the chain:* $\dot{y} = f(y, X_t)$, so the vector field jumps when the chain jumps, while the chain's rates may be constant or depend only on X.

(ii) *Chain driven by the ODE:* the transition rates $q_{ij}(y_t)$ of the chain depend on the continuous state, while $\dot{y} = f(y)$ evolves autonomously between jumps (or continuously).

(iii) *Two-way coupling:* both $f = f(y, X)$ and $q_{ij} = q_{ij}(y)$ are active, so neither component is autonomous.

Piecewise-deterministic Markov processes formalise (i)–(iii) when the continuous motion is deterministic between jumps (figure 2.8). The exponential-clock construction of section 2.2.2.1 still supplies the holding times once rates are frozen at the current (y, X); between jumps one integrates the ODE.

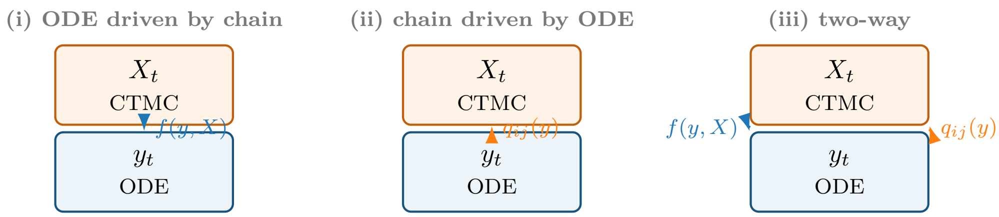


Figure 2.8: Coupled ODE–CTMC systems: dependence may run from the discrete state to the vector field (i), from the continuous state to the transition rates (ii), or in both directions (iii). Between jumps the continuous component follows an ODE; holding times remain exponential at the frozen rates.

2.3 Methods for Markov chains

2.3.1 Discrete-time methods

2.3.1.1 First-step analysis

A recurring discrete-time device, used for hitting probabilities and for the Galton–Watson extinction probability in section 2.2.1.2, is *first-step* (or one-step) analysis. Let $(X_n)_{n\geq 0}$ be a discrete-time Markov chain on a countable state space $\mathcal{S}$, and let $A \subset \mathcal{S}$ be a target set. Write $h_i = \mathbb{P}(X \text{ hits } A \mid X_0 = i)$. Conditioning on the first transition and using the Markov property yields the linear system

$$h_i = 1 \quad (i \in A), \qquad h_i = \sum_{j\in\mathcal{S}} P_{ij}\, h_j \quad (i \notin A), \tag{2.22}$$

where P is the one-step transition matrix. The same idea produces mean hitting times, with an inhomogeneous term $+1$ on the right-hand side for transient states. In continuous time the analogous identities come from the generator rather than from P; both versions appear repeatedly below.

Generating functions and functional iteration

For Galton–Watson processes the offspring probability generating function ϕ iterates to give the law of Z_n. Survival probabilities satisfy a closed nonlinear recursion such as (2.2). Functional iteration, first-step analysis, and—later—product and series manipulations of the survival amplitude $A(p)$ are the discrete-time methods used in this thesis. The amplitude theory, including the identification of $A(p)$ with a value of the Koenigs linearising function of the logistic map, is developed in full in Chapter [The constant $A(p)$]. The present chapter uses only the definition of $A(p)$ and the conditional-mean formula of section 2.3.2.

2.3.2 Continuous-time methods

Throughout this subsection the linear birth–death process of section 2.2.2.5 is the running example. The same pattern—generator to master equation to generating-function PDE to moments—applies to birth–death–catastrophe and absorption models.

Backward equations and the generator

For a continuous-time Markov chain with generator Q, the transition semigroup $P(t)$ satisfies the Kolmogorov backward and forward equations $\dot{P} = QP = PQ$. Applied to a bounded observable f, the function $u_i(t) = \mathbb{E}(f(X_t) \mid X_0 = i)$ solves $\dot{u} = Qu$ with $u(0) = f$. First-step reasoning in continuous time is the expansion of this equation over an infinitesimal holding time.

Mean population size

For linear birth–death rates $q_{n,n+1} = \lambda n$, $q_{n,n-1} = \mu n$, the mean $m(t) = \mathbb{E}(X_t)$ closes:

$$\dot{m} = (\lambda - \mu)m, \qquad m(t) = m(0)\,\mathrm{e}^{(\lambda-\mu)t}, \tag{2.23}$$

as in (2.17). Higher moments satisfy linear ODEs obtained by differentiating the generating-function PDE of (2.15).

Variance and higher moments

Differentiating the probability generating function (or the moment generating function) along the PDE yields a hierarchy of moment equations. For linear birth–death processes the continuous-time second-moment ODEs follow from (2.15) by the same route used for the mean. Explicit formulae are omitted here when they are not needed for later chapters.

Hitting probabilities

Let $A \subset \mathcal{S}$ be a target set for a continuous-time chain. The hitting probabilities $h_i = \mathbb{P}(\text{hit } A \mid X_0 = i)$ solve the linear system

$$\sum_j q_{ij} h_j = 0 \quad (i \notin A), \qquad h_i = 1 \quad (i \in A), \tag{2.24}$$

with the boundary condition $h = 0$ on any other absorbing class if required. For birth–death processes on an interval the system reduces to a second-order difference equation, solvable by summation exactly as in the discrete-time gambler's-ruin calculation.

Extinction probabilities

For continuous-time birth–death processes started at n, the ultimate extinction probability η_n solves a first-step balance using λ and μ; for the linear case, $\eta_1 = \min(1, \mu/\lambda)$. In discrete time the extinction probability of a Galton–Watson process is the smallest non-negative root of $s = \phi(s)$. Both facts are classical [3, 2].

Conditional means

For the subcritical binary Galton–Watson process ($p < \frac{1}{2}$), late-time survival obeys

$$S_n \sim A(p)\,(2p)^n, \qquad A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n}, \tag{2.25}$$

and the conditional mean converges to the reciprocal of this constant:

$$\mathbb{E}(Z_n \mid Z_n > 0) \xrightarrow{n\to\infty} \frac{1}{A(p)}. \tag{2.26}$$

Existence of the Yaglom (quasi-stationary) limit is classical [4, 5, 6]. The structure of $A(p)$ itself—product and series representations, bounds, near-critical asymptotics, and the obstruction to an elementary closed form—is the subject of Chapter [The constant $A(p)$].

Continuous-time closed form. The continuous-time linear birth–death process admits a closed form for its Kolmogorov constant, developed here in basic detail. By contrast, the discrete constant $A(p)$ has no known elementary formula; its full analytic theory is deferred to Chapter [The constant $A(p)$].

Take the linear birth–death process of section 2.2.2.5, with per-capita birth rate λ and death rate μ, started from N individuals. Its extinction probability $p_0(t)$ is (2.18), and its survival probability is $S^{(N)}(t) = 1 - p_0(t)$.

Limiting mean conditioned on non-extinction

The process has a stationary conditional mean when $\mu > \lambda$, that is in the subcritical case. Exactly as in section 2.3.2,

$$\mathbb{E}(X_t \mid X_t > 0) = \frac{\mathbb{E}(X_t)}{S^{(N)}(t)}, \tag{2.27}$$

and the late-time limit follows on applying L'Hôpital's rule. It is quicker to take $N = 1$ and compute directly. With $\mathbb{E}(X_t) = \mathrm{e}^{(\lambda-\mu)t}$ from (2.17) and

$$S(t) = 1 - \frac{\mu - \mu\mathrm{e}^{(\mu-\lambda)t}}{\lambda - \mu\mathrm{e}^{(\mu-\lambda)t}} = \frac{\lambda - \mu}{\lambda - \mu\mathrm{e}^{(\mu-\lambda)t}},$$

(2.27) gives

$$\mathbb{E}(X_t \mid X_t > 0) = \frac{\lambda\mathrm{e}^{(\lambda-\mu)t} - \mu}{\lambda - \mu}.$$

For $\mu > \lambda$ the exponential decays, and the limiting conditional mean is the constant

$$\lim_{t\to\infty} \mathbb{E}(X_t \mid X_t > 0) = \frac{\mu}{\mu - \lambda} = \frac{1}{1 - \lambda/\mu} \left(= \frac{1}{A_\mathrm{c}} \right). \tag{2.28}$$

Figure 2.9 shows the approach of the conditional mean to this limit for a representative subcritical pair (λ, μ). To compare this with the discrete-time constant the two parametrisations must be matched, and proposition 2.4 is what matches them: with birth and death clocks running at rates λ and μ, the probability that the birth clock wins is $p = \lambda/(\lambda + \mu)$, so that $\lambda/\mu = p/(1-p)$. In that parametrisation

$$\frac{1}{A_\mathrm{c}} = \frac{1-p}{1-2p}, \qquad \text{that is} \qquad A_\mathrm{c}(p) = \frac{1-2p}{1-p}, \tag{2.29}$$

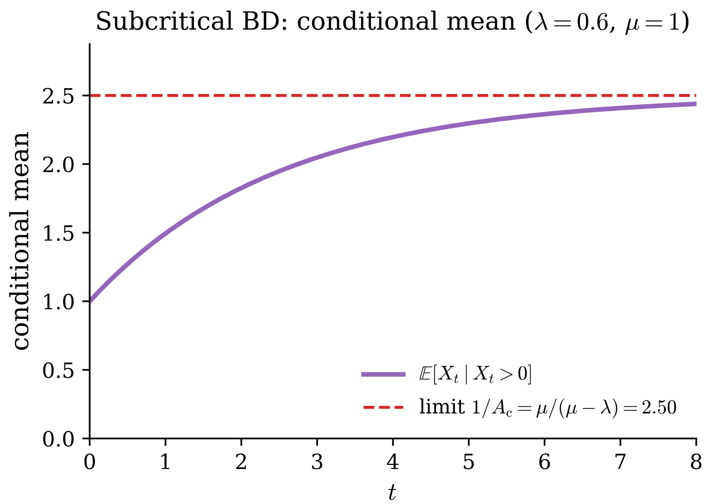


Figure 2.9: Subcritical linear birth–death process: the conditional mean $\mathbb{E}(X_t \mid X_t > 0)$ rises from 1 to the stationary value $1/A_\mathrm{c} = \mu/(\mu - \lambda)$ (dashed). Parameters $\lambda = 0.6$, $\mu = 1$.

the subscript marking this as the continuous-time constant, to distinguish it from the discrete-time $A(p)$ of (2.25).

Figure 2.10 plots $A_\mathrm{c}(p)$ against numerically obtained values of $A(p)$ and raises a natural question: why does the continuous curve start at 1 and the discrete one at $\frac{1}{2}$? Merely halving A_c does not superpose them, because at the other end of the range they meet. As shown in Chapter [The constant $A(p)$], $A_\mathrm{c}(p)/2$ is an exact lower bound for $A(p)$, attained only as $p \to 0$, while near criticality both constants share the leading asymptotic $2(1 - 2p)$. The product, series, bounds, and dynamical obstruction for discrete $A(p)$ occupy Chapter [The constant $A(p)$].

The continuous-time constant is elementary because the generating-function PDE closes. The discrete constant $A(p)$ is not known to admit an elementary formula; that contrast, and the full discrete theory, occupy Chapter [The constant $A(p)$].

2.3.3 Method of characteristics

Generating-function equations for continuous-time population models are first-order linear PDEs. The method of characteristics converts such a PDE into a system of ordinary differential equations along curves in the (t, x) (or multi-variable) space (figure 2.11). The presentation below is organised for readability: the method is checked on the absorption–death model, whose answer is already known by an independence argument, and the longer absorption–birth–death closed form is recorded in section 2.A.

2.3.3.1 Worked example: absorption–death

A particle resident in a compartment may die. Keeping X_t and Y_t for the interior and exterior counts, the two channels are absorption from the exterior medium into the compartment at per-particle rate α, and death inside at per-particle rate μ (figure 2.12). The infinitesimal rates read

$$\mathbb{P}(\Delta X = 1,\ \Delta Y = -1 \mid Y_t = j) = j\alpha\,\Delta t, \qquad \text{(absorption)}$$
$$\mathbb{P}(\Delta X = -1,\ \Delta Y = 0 \mid X_t = i) = i\mu\,\Delta t, \qquad \text{(death)}$$

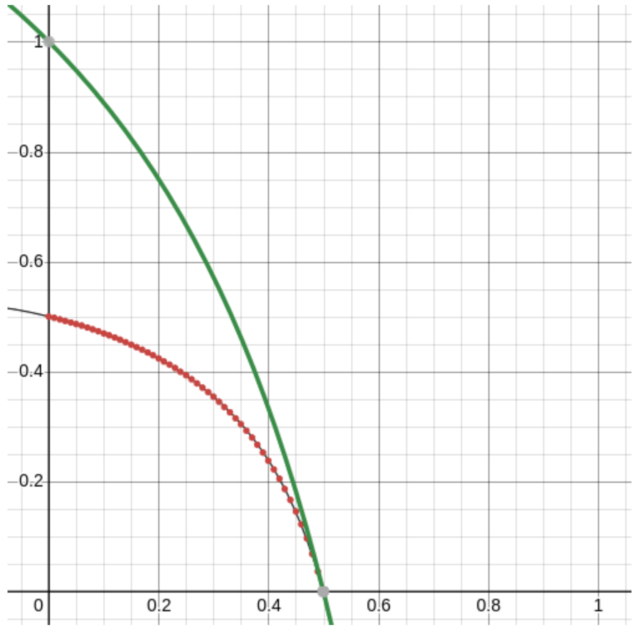


Figure 2.10: The continuous-time constant $A_c(p) = (1-2p)/(1-p)$ (green) against high-precision numerical values of the discrete-time constant $A(p)$ (red points). Both vanish at $p = \frac{1}{2}$; at $p = 0$ the continuous curve starts at 1 and the discrete one at $\frac{1}{2}$, and the two are not related by a constant factor in between. The black line is a symbolic-regression candidate discussed in Chapter [The constant $A(p)$]; it tracks the points closely but is not exact. The exact relation between A and A_c is established in Chapter [The constant $A(p)$].

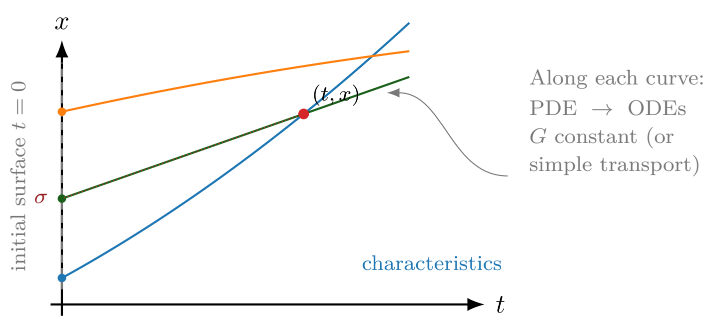


Figure 2.11: Schematic of the method of characteristics for a first-order linear PDE in (t, x). Solution values at a point (t, x) are recovered by integrating the characteristic ODEs back to the initial surface $t = 0$, where G is known. In the absorption models the same idea is applied in (t, x, y) with a two-dimensional initial surface.

$$\mathbb{P}(\Delta X = 0,\ \Delta Y = 0 \mid X_t = i,\ Y_t = j) = 1 - (j\alpha + i\mu)\Delta t, \qquad \text{(no event)}$$

with forward Kolmogorov equations

$$\frac{\mathrm{d}p_{i,j}}{\mathrm{d}t} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} - (\alpha j + \mu i)p_{i,j}. \tag{2.30}$$

The derivation of section 2.A.1 carries over unchanged and gives the first-order linear equation

$$\frac{\partial G}{\partial t} = \mu(1-x)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}, \qquad G(x, y, 0) = y^N, \tag{2.31}$$

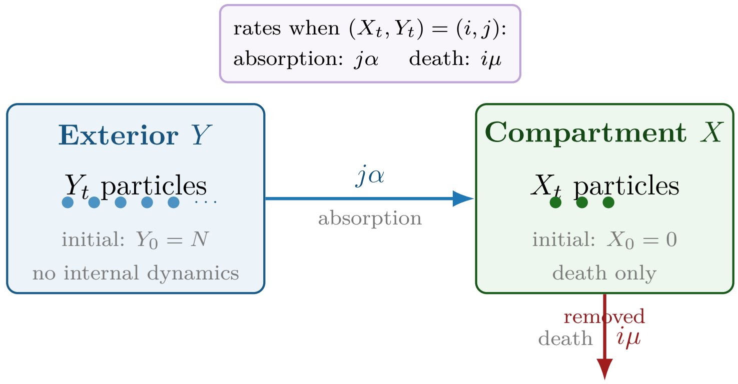


Figure 2.12: Absorption–death compartment schematic. Particles start in the exterior medium Y (count Y_t, initially N). Each exterior particle is absorbed into the compartment X at rate α, contributing the total absorption rate $j\alpha$ when $Y_t = j$. Inside X, each of the i residents dies independently at rate μ, for total death rate $i\mu$. There is no birth in this model.

and the same difficulty. Once again the solution can be constructed via (2.46), since individual particles remain independent.

Without birth events the number of particles cannot exceed its initial value, so for an isolated particle the generating function has just three terms,

$$G_1(x, y, t) = p_{0,0}(t) + p_{1,0}(t)\, x + p_{0,1}(t)\, y.$$

The state probabilities follow from

$$\frac{\mathrm{d}p_{0,1}}{\mathrm{d}t} = -\alpha p_{0,1}, \qquad \frac{\mathrm{d}p_{1,0}}{\mathrm{d}t} = \alpha p_{0,1} - \mu p_{1,0},$$

whence

$$p_{0,1}(t) = \mathrm{e}^{-\alpha t}, \qquad p_{1,0}(t) = \frac{\alpha}{\mu - \alpha}\left(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\mu t}\right), \qquad p_{0,0}(t) = 1 - p_{0,1}(t) - p_{1,0}(t), \tag{2.32}$$

plotted in figure 2.13. Applying (2.46) gives the general generating function

$$G_n(x, y, t) = \left(1 - \tfrac{\mu}{\mu-\alpha}\mathrm{e}^{-\alpha t} + \tfrac{\alpha}{\mu-\alpha}\mathrm{e}^{-\mu t} + \tfrac{\alpha}{\mu-\alpha}(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\mu t})x + \mathrm{e}^{-\alpha t}y\right)^n. \tag{2.33}$$

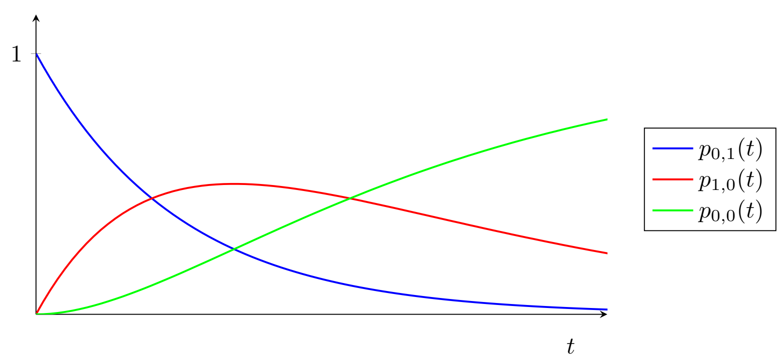


Figure 2.13: State probabilities for the single-particle absorption–death model, (2.32). The particle starts outside ($p_{0,1}$, blue), is absorbed into the compartment ($p_{1,0}$, red, which rises and then decays as death takes over), and finally dies ($p_{0,0}$, green, absorbing).

2.3.3.2 Characteristics applied to absorption–death

Before continuing to absorption–birth–death, where replication may occur inside the compartment and the independence shortcut is unavailable, it is worth revisiting absorption–death from the point of view of its partial differential equation. Having (2.33) already in hand makes it an ideal test case.[1]

Introduce a change of variables $(x, y, t) \to (\sigma, \eta, \tau)$, writing

$$G(x, y, t) = G\big(x(\sigma, \eta, \tau),\ y(\sigma, \eta, \tau),\ t(\sigma, \eta, \tau)\big) \equiv G(\sigma, \eta, \tau).$$

The method seeks solutions along the characteristic curves of constant σ and η, on which G is a function of τ alone. To fix the parametrisation we impose an initial surface,

$$x(\sigma, \eta, 0) = \sigma, \qquad y(\sigma, \eta, 0) = \eta, \qquad t(\sigma, \eta, 0) = 0. \tag{2.34}$$

With G a function of τ only along a characteristic, the chain rule gives

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = \frac{\partial G}{\partial t}\frac{\mathrm{d}t}{\mathrm{d}\tau} + \frac{\partial G}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}\tau} + \frac{\partial G}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}\tau},$$

which is to be compared with (2.31) written as

$$0 = -\frac{\partial G}{\partial t} + \mu(1 - x)\frac{\partial G}{\partial x} + \alpha(x - y)\frac{\partial G}{\partial y}.$$

Comparing coefficients yields the characteristic system

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = 0, \tag{2.35}$$
$$\frac{\mathrm{d}t}{\mathrm{d}\tau} = -1, \tag{2.36}$$
$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = \mu(1 - x), \tag{2.37}$$
$$\frac{\mathrm{d}y}{\mathrm{d}\tau} = \alpha(x - y). \tag{2.38}$$

The equation for G, (2.35). G is constant along characteristics, so G is a function of (σ, η) alone. Under (2.34) the initial condition $G(x, y, 0) = y^N$ becomes $G(\sigma, \eta, 0) = \eta^N$, so

$$G(\sigma, \eta, \tau) = \eta^N. \tag{2.39}$$

The whole problem thus reduces to expressing η in terms of (x, y, t).

The equation for t, (2.36). Integrating gives $t = -\tau + c(\sigma, \eta)$, and (2.34) fixes $c = 0$:

$$t = -\tau. \tag{2.40}$$

The characteristic parameter runs backwards in time, which is what allows the initial surface to be reached from any (x, y, t).

[1] I am grateful to Benjamin Sharp for the advice on applying the method of characteristics to three-variate problems that unlocked this section.

The equation for x, (2.37). Separating variables gives $1 - x = c(\sigma, \eta)\mathrm{e}^{-\mu\tau}$, and (2.34) gives $c = 1 - \sigma$:

$$1 - x = (1 - \sigma)\mathrm{e}^{-\mu\tau}. \tag{2.41}$$

The equation for y, (2.38). Substituting (2.41),

$$\frac{\mathrm{d}y}{\mathrm{d}\tau} = \alpha(1 - (1 - \sigma)\mathrm{e}^{-\mu\tau} - y),$$

and solving with integrating factor $\mathrm{e}^{\alpha\tau}$,

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1 - \sigma)\mathrm{e}^{-\mu\tau} + c(\sigma, \eta)\mathrm{e}^{-\alpha\tau}.$$

Applying (2.34) gives $c(\sigma, \eta) = \eta + \frac{\alpha}{\alpha-\mu}(1 - \sigma) - 1$, so

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1 - \sigma)\mathrm{e}^{-\mu\tau} + \left(\eta + \frac{\alpha}{\alpha - \mu}(1 - \sigma) - 1\right)\mathrm{e}^{-\alpha\tau}. \tag{2.42}$$

Recovering the solution. Rearranging (2.42) for η, substituting σ from (2.41) and $\tau = -t$ from (2.40), gives

$$\eta = 1 - \frac{\mu}{\mu - \alpha}\mathrm{e}^{-\alpha t} + \frac{\alpha}{\mu - \alpha}\mathrm{e}^{-\mu t} + \frac{\alpha}{\mu - \alpha}(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\mu t})x + \mathrm{e}^{-\alpha t}y,$$

and (2.39) then reproduces (2.33) exactly. The two routes agree, and the machinery is ready for the case in which only one of them is available.

Remark 2.11 (Further absorption models). Absorption-only and absorption–birth–death generating functions, including the hypergeometric closed form and resonance discussion, are collected in section 2.A. The hypergeometric integral identity and coefficient extraction lemmas remain in sections 2.B and 2.C.

2.A Extended absorption models

This appendix records the absorption-only and absorption–birth–death developments that support section 2.3.3 but are not required for the single worked example in the main text.

2.A.1 Absorption only

The simplest absorption process has a single compartment that takes in particles, with no subsequent dynamics: no birth and no death within the compartment.

Let X_t and Y_t count the particles inside and outside the compartment respectively; the same labelling is used for every version of the single-compartment model below. Events are stochastic with exponentially distributed inter-event times, which by section 2.2.2.1 follows from the assumption that event rates do not depend on time, so that $\{(X_t, Y_t) : t \geq 0\}$ is a continuous-time Markov chain in the sense of section 2.2.2.2.

With only absorption to consider, the model is specified by the transition probabilities over an infinitesimal interval Δt,

$$\mathbb{P}(\Delta X = 1,\ \Delta Y = -1 \mid Y_t = j) = j\alpha\,\Delta t, \qquad \text{(absorption)}$$

$$\mathbb{P}(\Delta X = 0,\ \Delta Y = 0 \mid Y_t = j) = 1 - j\alpha\,\Delta t, \qquad \text{(no event)}$$

where $\Delta X = X_{t+\Delta t} - X_t$ and similarly for Y, and terms of $o(\Delta t)$ are suppressed throughout. The factor j records that each exterior particle is independently subject to the same probability $\alpha\Delta t$ of being taken up. The initial condition is $X_0 = 0$, $Y_0 = N$: a starter population of N particles occupying the exterior medium.

Writing $p_{i,j}(t) = \mathbb{P}(X_t = i,\ Y_t = j)$, the forward Kolmogorov equations are

$$\frac{\mathrm{d}p_{i,j}}{\mathrm{d}t} = \alpha(j+1)p_{i-1,j+1} - \alpha j\, p_{i,j}. \tag{2.43}$$

With the bivariate generating function

$$G(x,y,t) = \sum_i \sum_j p_{i,j}(t)\, x^i y^j = \mathbb{E}\Big(x^{X_t} y^{Y_t}\Big), \tag{2.44}$$

(2.43) converts into a partial differential equation by the procedure of section 2.2.2.3. Multiply each equation by $x^i y^j$ and sum over i and j; the sums are finite, because $p_{i,j}(t) = 0$ whenever $i + j \neq N$, particles in this model being neither created nor destroyed. Shifting indices in the first term sends $\alpha(j+1)p_{i-1,j+1}x^i y^j$ to $\alpha x\,\partial_y G$, and the second term is $-\alpha y\,\partial_y G$, so that

$$\frac{\partial G}{\partial t} = \alpha(x-y)\frac{\partial G}{\partial y}, \qquad G(x,y,0) = y^N. \tag{2.45}$$

This first-order linear initial value problem is more awkward than it looks. The method of characteristics applies in principle to equations in three variables, but applying it here is not straightforward, and it is postponed to section 2.3.3.2. The solution can be obtained by other means, and it is worth doing so both because the answer is needed and because it provides a check on the machinery developed there.

Exploiting independence

The function G above was specific to $X_0 = 0$, $Y_0 = N$. Define more generally

$$G_n(x,y,t) = \sum_i \sum_j \mathbb{P}(X_t = i,\ Y_t = j \mid X_0 = 0,\ Y_0 = n)\ x^i y^j.$$

In a process of pure absorption the n particles are taken up one by one, and the behaviour of any one of them does not depend on the movements of its neighbours. That independence lets X_t and Y_t decompose into sums of n single-particle variables,

$$X_t = \sum_{k=1}^{n} X_t^{(k)}, \qquad Y_t = \sum_{k=1}^{n} Y_t^{(k)},$$

from which

$$G_n(x,y,t) = \big(G_1(x,y,t)\big)^n, \tag{2.46}$$

since

$$G_n(x,y,t) = \mathbb{E}\Bigg(\prod_{k=1}^{n} x^{X_t^{(k)}} \prod_{k=1}^{n} y^{Y_t^{(k)}}\Bigg) = \prod_{k=1}^{n} \mathbb{E}\Big(x^{X_t^{(k)}} y^{Y_t^{(k)}}\Big) = \big(G_1(x,y,t)\big)^n.$$

Now consider a single initial particle, $X_0 = 0$, $Y_0 = 1$. It is eventually absorbed, and that is the final state of the system: no other states are reachable, so $p_{i,j}(t) = 0$ whenever $i + j \neq 1$. The forward equations reduce to

$$\frac{\mathrm{d}p_{0,1}}{\mathrm{d}t} = -\alpha p_{0,1}, \qquad \frac{\mathrm{d}p_{1,0}}{\mathrm{d}t} = \alpha p_{0,1},$$

giving $p_{0,1}(t) = \mathrm{e}^{-\alpha t}$ and $p_{1,0}(t) = 1 - \mathrm{e}^{-\alpha t}$ (figure 2.14). These being the only two non-zero state probabilities of the single-particle system,

$$G_1(x, y, t) = p_{1,0}(t)\, x + p_{0,1}(t)\, y = \left(1 - \mathrm{e}^{-\alpha t}\right) x + \mathrm{e}^{-\alpha t} y,$$

and (2.46) gives the generating function for any initial census,

$$G_n(x, y, t) = \left(\left(1 - \mathrm{e}^{-\alpha t}\right) x + \mathrm{e}^{-\alpha t} y\right)^n. \tag{2.47}$$

This satisfies (2.45) together with the initial condition $G_n(x, y, 0) = y^n$. The form is recognisable: $X_t \sim \text{Binomial}(n, 1 - \mathrm{e}^{-\alpha t})$, as it must be, since the n particles are absorbed independently and each has been taken up by time t with probability $1 - \mathrm{e}^{-\alpha t}$.

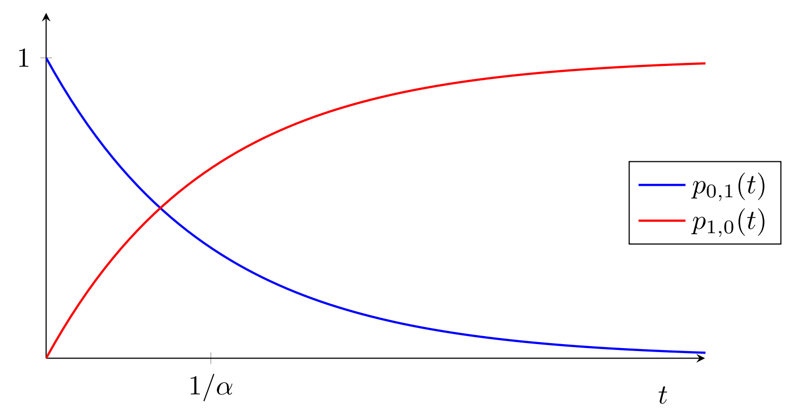


Figure 2.14: State probabilities for the single-particle absorption-only model. The particle begins outside the compartment ($p_{0,1}$, decaying exponentially at rate α) and ends inside it ($p_{1,0}$). There is nowhere else for it to go, so the two curves sum to one at every time.

2.A.2 Absorption–birth–death

Introducing replication for particles inside the compartment, the independence property is still present but no longer serves as a shortcut: from a single initial particle, birth events make infinitely many states reachable, so $G_1(x, y, t)$ is an infinite sum and there is nothing to be gained by factorising. The partial differential equation must therefore be solved directly, and by now that can be done.

With X_t and Y_t as before, birth at per-capita rate λ is added to the transition probabilities:

$$\begin{aligned}
\mathbb{P}(\Delta X = 1,\ \Delta Y = -1 \mid Y_t = j) &= j\alpha\,\Delta t, && \text{(absorption)}\\
\mathbb{P}(\Delta X = 1,\ \Delta Y = 0 \mid X_t = i) &= i\lambda\,\Delta t, && \text{(birth)}\\
\mathbb{P}(\Delta X = -1,\ \Delta Y = 0 \mid X_t = i) &= i\mu\,\Delta t, && \text{(death)}\\
\mathbb{P}(\Delta X = 0,\ \Delta Y = 0 \mid X_t = i,\ Y_t = j) &= 1 - (j\alpha + i\mu + i\lambda)\Delta t, && \text{(no event)}
\end{aligned}$$

with forward Kolmogorov equation

$$\frac{\mathrm{d}p_{i,j}}{\mathrm{d}t} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} + \lambda(i-1)p_{i-1,j} - (\alpha j + \mu i + \lambda i)p_{i,j}. \tag{2.48}$$

The same reasoning as before gives the generating-function equation

$$\frac{\partial G}{\partial t} = \left(\mu - (\mu+\lambda)x + \lambda x^2\right)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}, \qquad G(x,y,0) = y^N, \tag{2.49}$$

whose x-coefficient is exactly the quadratic met in (2.15) for the linear birth–death process. Reparametrising $(x,y,t) \to (\sigma,\eta,\tau)$ as in section 2.3.3.2 gives the characteristic system

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = 0, \tag{2.50}$$

$$\frac{\mathrm{d}t}{\mathrm{d}\tau} = -1, \tag{2.51}$$

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = \mu - (\mu+\lambda)x + \lambda x^2, \tag{2.52}$$

$$\frac{\mathrm{d}y}{\mathrm{d}\tau} = \alpha(x-y), \tag{2.53}$$

with the same initial conditions (2.34).

The equations for G and t, (2.50)–(2.51). As before,

$$G(\sigma,\eta,\tau) = \eta^N, \qquad t = -\tau. \tag{2.54}$$

The equation for x, (2.52). This is a Riccati equation of the same shape as the one governing the survival probability in the continuum approximation of Chapter [The constant $A(p)$]. Its right-hand side factorises, and the roots are elementary rather than the surd expression that mechanical use of the quadratic formula suggests:

$$\mu - (\mu+\lambda)x + \lambda x^2 = \lambda(x-x_+)(x-x_-), \qquad x_+ = 1, \quad x_- = \frac{\mu}{\lambda}. \tag{2.55}$$

The discriminant is $\left(\frac{\mu+\lambda}{2\lambda}\right)^2 - \frac{\mu}{\lambda} = \left(\frac{\lambda-\mu}{2\lambda}\right)^2$, a perfect square, so the surds cancel. It is convenient to name the combination that appears throughout,

$$b_1 = \lambda(x_+ - x_-) = \lambda - \mu, \tag{2.56}$$

the net interior growth rate. Separating variables in (2.52) and applying (2.34) gives the characteristic in the compact form

$$\frac{x-x_+}{x-x_-} = \left(\frac{\sigma-x_+}{\sigma-x_-}\right)\mathrm{e}^{b_1\tau}, \tag{2.57}$$

or, solved for x,

$$x(\tau) = \frac{x_+(\sigma-x_-) - x_-(\sigma-x_+)\mathrm{e}^{b_1\tau}}{(\sigma-x_-) - (\sigma-x_+)\mathrm{e}^{b_1\tau}}. \tag{2.58}$$

The equation for y, (2.53). The last characteristic equation is $\mathrm{d}y/\mathrm{d}\tau = \alpha(x-y)$, which with an integrating factor becomes

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(\mathrm{e}^{\alpha\tau}y\right) = \alpha\,\mathrm{e}^{\alpha\tau}x(\tau). \tag{2.59}$$

The integral on the right is where the difficulty lies. Rather than substitute (2.58) and split the resulting quotient — which leads to two hypergeometric integrals and a good deal of bookkeeping — it is cleaner to expand $x(\tau)$ first. Writing $x - x_- = (x - x_+) + (x_+ - x_-)$ in (2.57) and solving for x gives

$$x(\tau) = x_+ + (x_+ - x_-)\frac{\zeta(\tau)}{1-\zeta(\tau)}, \qquad \zeta(\tau) = \left(\frac{\sigma - x_+}{\sigma - x_-}\right)\mathrm{e}^{b_1\tau}, \tag{2.60}$$

so that, for $|\zeta| < 1$,

$$x(\tau) = x_+ + (x_+ - x_-)\sum_{n=1}^{\infty}\zeta(\tau)^n.$$

Every term is now a pure exponential in τ and integrates immediately. Writing $Z = (\sigma - x_+)/(\sigma - x_-)$, so that $\zeta(\tau) = Z\mathrm{e}^{b_1\tau}$ and $Z^n\mathrm{e}^{nb_1\tau} = \zeta^n$,

$$\begin{aligned}\alpha\int \mathrm{e}^{\alpha\tau}x(\tau)\,\mathrm{d}\tau &= \alpha x_+\int \mathrm{e}^{\alpha\tau}\mathrm{d}\tau + \alpha(x_+ - x_-)\sum_{n=1}^{\infty}Z^n\int \mathrm{e}^{(\alpha+nb_1)\tau}\mathrm{d}\tau \\ &= \mathrm{e}^{\alpha\tau}\left[x_+ + \alpha(x_+ - x_-)\sum_{n=1}^{\infty}\frac{\zeta^n}{\alpha + nb_1}\right] + \text{const.}\end{aligned}$$

Setting $\delta = \alpha/b_1$ and using $\alpha(x_+ - x_-)/b_1 = \alpha/\lambda$ by (2.56), this defines the function that carries the whole solution:

$$\Psi(\zeta) = x_+ + \frac{\alpha}{\lambda}\sum_{n=1}^{\infty}\frac{\zeta^n}{n+\delta}, \qquad \delta = \frac{\alpha}{b_1} = \frac{\alpha}{\lambda - \mu}. \tag{2.61}$$

So $\alpha\int \mathrm{e}^{\alpha\tau}x(\tau)\mathrm{d}\tau = \mathrm{e}^{\alpha\tau}\Psi(\zeta(\tau)) + \text{const}$, and integrating (2.59) gives $\mathrm{e}^{\alpha\tau}y = \mathrm{e}^{\alpha\tau}\Psi(\zeta(\tau)) + \text{const}$. The initial condition $y(\sigma,\eta,0) = \eta$, at which $\zeta(0) = Z$, fixes the constant as $\eta - \Psi(Z)$, so

$$\eta = \mathrm{e}^{\alpha\tau}\Big[y - \Psi\big(\zeta(\tau)\big)\Big] + \Psi(Z). \tag{2.62}$$

Finally, put $\tau = -t$ and use (2.57): writing

$$W = \frac{x - x_+}{x - x_-}$$

we have $\zeta(\tau) = W$ and $Z = W\mathrm{e}^{-b_1\tau} = W\mathrm{e}^{b_1 t}$. Substituting into (2.62) and applying (2.54) gives the solution.

Proposition 2.12 (Generating function for absorption–birth–death)**.** *With $x_+ = 1$, $x_- = \mu/\lambda$, $b_1 = \lambda - \mu$, $\delta = \alpha/b_1$, $W = (x - x_+)/(x - x_-)$ and Ψ as in (2.61),*

$$\boxed{G(x,y,t) = \Big\{\mathrm{e}^{-\alpha t}\big[y - \Psi(W)\big] + \Psi\big(W\mathrm{e}^{b_1 t}\big)\Big\}^N.} \tag{2.63}$$

Three checks confirm the result. At $t = 0$ the two Ψ terms cancel and $G = y^N$, the required initial condition. At $x = y = 1$ we have $W = 0$ and $\Psi(0) = x_+ = 1$, so $G = 1$ and probability is conserved. And setting $x = 1$ alone gives

$$G(1,y,t) = \left(\mathrm{e}^{-\alpha t}y + 1 - \mathrm{e}^{-\alpha t}\right)^N,$$

so $Y_t \sim \mathrm{Binomial}(N, \mathrm{e}^{-\alpha t})$ — exactly right, since the exterior particles are absorbed independently at rate α and what happens inside the compartment cannot affect them. Beyond these, (2.63) has been verified numerically against direct integration of the master equation (2.48), agreeing to around 10^{-13} across sub- and supercritical interior regimes and for $N = 1, 2, 3$.

Hypergeometric form

The series in (2.61) is a Gauss hypergeometric function in disguise. With

$$ {}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!}, \qquad (q)_n = \begin{cases} 1, & n = 0, \\ q(q+1)\cdots(q+n-1), & n \geq 1, \end{cases} \tag{2.64} $$

where $(q)_n$ is the rising Pochhammer symbol [7], the case $a = 1$, $b = \delta$, $c = \delta + 1$ collapses to a single ratio. Since $(1)_n = n!$ and

$$ \frac{(\delta)_n}{(\delta+1)_n} = \frac{\delta(\delta+1)\cdots(\delta+n-1)}{(\delta+1)\cdots(\delta+n-1)(\delta+n)} = \frac{\delta}{\delta+n}, \tag{2.65} $$

we have

$$ {}_2F_1(1, \delta; \delta+1; z) = \sum_{n=0}^{\infty} \frac{\delta}{\delta+n} z^n. \tag{2.66} $$

Comparing with (2.61) and using $\alpha/(\lambda\delta) = b_1/\lambda = x_+ - x_-$,

$$ \Psi(\zeta) = x_+ + (x_+ - x_-)\Big[{}_2F_1(1, \delta; \delta+1; \zeta) - 1\Big]. \tag{2.67} $$

(2.63) together with (2.67) is the closed form. It is not pretty, but it is a genuine closed form in terms of a standard special function, and the elementary series (2.61) is what one would actually evaluate. An alternative derivation of the same object, by way of the integral $\int \mathrm{e}^{h\tau}/(P + Q\mathrm{e}^{k\tau})\,\mathrm{d}\tau$, is given in section 2.B; that route is longer, but it produces a general identity of independent use.

Remark 2.13 (Domain of validity). The series (2.61) converges for $|\zeta| < 1$, and the arguments appearing in (2.63) are W and $W\mathrm{e}^{b_1 t}$. In the subcritical interior regime $\mu > \lambda$ we have $b_1 < 0$ and $x_- = \mu/\lambda > 1$, so for $x \in [0, 1]$ the argument W lies in $[0, \lambda/\mu] \subset [0, 1)$ and $W\mathrm{e}^{b_1 t}$ is smaller still: the series representation is valid on the whole of the unit square, which is the case of most interest here. When $\lambda > \mu$ the series converges only for x above the smaller fixed point μ/λ, and elsewhere one must use the analytic continuation of ${}_2F_1$ — conveniently, the Euler integral

$$ {}_2F_1(1, \delta; \delta+1; \zeta) = \delta \int_0^1 \frac{u^{\delta-1}}{1 - \zeta u}\,\mathrm{d}u \qquad (\delta > 0,\ \zeta \notin [1, \infty)), $$

which was used for the numerical verification quoted above.

Remark 2.14 (Resonance). The denominators in (2.61) vanish when $n + \delta = 0$ for some integer $n \geq 1$, that is when

$$ \alpha = n(\mu - \lambda), \qquad n = 1, 2, \ldots $$

This is the classical resonance condition for linearising the y-equation: the decay rate α of the absorption channel coincides with a harmonic of the interior relaxation rate $\mu - \lambda$. The generating function itself is perfectly regular there, being analytic in all parameters, so the singularity is removable, and at resonance the corresponding term acquires a factor τ — a secular term — in place of the simple exponential. The resonant parameter values form a set of measure zero and the limit may be taken numerically, but the degeneracy should be kept in mind when α happens to sit near an integer multiple of $\mu - \lambda$.

Parameter-regular form and resonance

The closed form (2.63) can be rewritten so that the apparent singularity at resonant parameter values is removable. When the hypergeometric parameters meet a non-positive integer resonance, the series representation degenerates and the correct finite limit must be taken separately; the underlying characteristic solution remains regular. In practice one evaluates the parameter-regular expression (or takes the resonant limit of the hypergeometric form) rather than trusting a naïve specialisation of intermediate formulae. Domains of the hypergeometric representation and the resonant cases are recorded in the source derivations; the identities of section 2.B cover the integral manipulations used above.

2.B An integral identity for the hypergeometric function

The route to (2.61) taken in section 2.A.2 expands the characteristic $x(\tau)$ in a geometric series and integrates term by term. An alternative is to integrate the quotient (2.58) directly, splitting the numerator into two pieces, each of the form

$$I(\tau) = \int \frac{\mathrm{e}^{h\tau}}{P + Q\,\mathrm{e}^{k\tau}}\,\mathrm{d}\tau \tag{2.68}$$

with h, k, P, Q constant. That route is longer and easier to get wrong — the two pieces carry different prefactors, and dropping one produces a formula that still satisfies the initial condition while violating $G(1,1,t) = 1$ — but the identity it rests on is useful in its own right, so it is recorded and proved here.

Proposition 2.15. *For $h/k > 0$ and $-\frac{Q}{P}\mathrm{e}^{k\tau} \notin [1,\infty)$,*

$$\int \frac{\mathrm{e}^{h\tau}}{P + Q\mathrm{e}^{k\tau}}\,\mathrm{d}\tau = \frac{\mathrm{e}^{h\tau}}{Ph}\,{}_2F_1\left(1, \frac{h}{k}; \frac{h}{k}+1; -\frac{Q}{P}\mathrm{e}^{k\tau}\right) + \mathit{const.} \tag{2.69}$$

Proof. We use the Euler integral representation of the hypergeometric function [7],

$${}_2F_1(a,b;c;z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a}\,\mathrm{d}u, \tag{2.70}$$

Γ being the gamma function.

Start from (2.68) and substitute $v = \mathrm{e}^{k\tau}$, so that $\mathrm{d}\tau = k^{-1}v^{-1}\mathrm{d}v$:

$$I = \frac{1}{k}\int \frac{v^{h/k}}{P + Qv}\,v^{-1}\mathrm{d}v = \frac{1}{Pk}\int \frac{v^{h/k-1}}{1+\frac{Q}{P}v}\,\mathrm{d}v.$$

Fix the antiderivative by writing this as a definite integral from 0 to v in a dummy variable w; the integral converges at the lower limit precisely because $h/k > 0$:

$$I = \frac{1}{Pk}\int_0^v \frac{w^{h/k-1}}{1+\frac{Q}{P}w}\,\mathrm{d}w.$$

Now rescale, $w = vu$, so that $\mathrm{d}w = v\,\mathrm{d}u$ with v held fixed:

$$I = \frac{1}{Pk}\int_0^1 \frac{(vu)^{h/k-1}}{1+\frac{Q}{P}vu}\,v\,\mathrm{d}u = \frac{v^{h/k}}{Pk}\int_0^1 \frac{u^{h/k-1}}{1+\frac{Q}{P}vu}\,\mathrm{d}u. \tag{2.71}$$

The remaining integrand matches that of (2.70) with $b = h/k$, $c = h/k + 1$, $a = 1$ and $z = -\frac{Q}{P}v$, since then $(1-u)^{c-b-1} = 1$ and $(1-zu)^{-a} = (1 + \frac{Q}{P}vu)^{-1}$. Hence, using $\Gamma(z+1) = z\Gamma(z)$,

$$\int_0^1 \frac{u^{h/k-1}}{1+\frac{Q}{P}vu}\,\mathrm{d}u = \frac{\Gamma\left(\frac{h}{k}\right)\Gamma(1)}{\Gamma\left(\frac{h}{k}+1\right)}\,{}_2F_1\left(1, \frac{h}{k}; \frac{h}{k}+1; -\frac{Q}{P}v\right) = \frac{k}{h}\,{}_2F_1\left(1, \frac{h}{k}; \frac{h}{k}+1; -\frac{Q}{P}v\right).$$

Substituting into (2.71) gives

$$I = \frac{v^{h/k}}{Ph}\,{}_2F_1\left(1, \frac{h}{k}; \frac{h}{k}+1; -\frac{Q}{P}v\right),$$

and restoring $v = \mathrm{e}^{k\tau}$ gives (2.69). □

Remark 2.16. Applied to the two pieces of (2.58) — the first with $h = \alpha$, the second with $h = \alpha + b_1$, and both with $k = b_1$, $P = \sigma - x_-$ and $Q = -(\sigma - x_+)$ — proposition 2.15 reproduces (2.67) once the prefactors $\alpha x_+(\sigma - x_-)$ and $-\alpha x_-(\sigma - x_+)$ are carried through correctly. The factor $(\sigma - x_+)/(\sigma - x_-)$ attached to the second piece is easily lost, and its loss is invisible at $t = 0$; checking $G(1,1,t) = 1$ catches it immediately.

2.C Extracting state probabilities from a generating function

A generating function defined by a formula rather than as an explicit series — for example

$$G(x,y,t) = \left((1-\mathrm{e}^{-\alpha t})x + \mathrm{e}^{-\alpha t}y\right)^n \tag{2.72}$$

— can always be converted back to the form

$$G(x,y,t) = \sum_{i=0}^{\infty}\sum_{j=0}^{\infty} p_{i,j}(t)\,x^i y^j, \tag{2.73}$$

recovering the time-dependent state probabilities. All that is required is that G be repeatedly partially differentiable in x and y.

The reason is the bivariate Maclaurin expansion,

$$f(x,y) = \sum_{k=0}^{\infty}\frac{1}{k!}\sum_{\varrho=0}^{k}\binom{k}{\varrho}\partial_x^{\varrho}\partial_y^{k-\varrho}f\Big|_{(0,0)}\,x^{\varrho}y^{k-\varrho},$$

whose coefficient of $x^i y^j$, taking $k = i+j$ and $\varrho = i$, is $\frac{1}{(i+j)!}\binom{i+j}{i}\partial_x^i\partial_y^j f\big|_{(0,0)}$. Hence

$$p_{i,j}(t) = \frac{1}{i!\,j!}\,\partial_x^i\partial_y^j G\Big|_{(0,0)}. \tag{2.74}$$

The multilinear case

Both (2.47) and (2.33) have the form

$$G(x,y,t) = \left(c(t) + f(t)\,x + g(t)\,y\right)^n, \tag{2.75}$$

for which the derivatives are immediate. Each differentiation lowers the power by one and brings down a factor, so

$$\partial_x^a \partial_y^b G = \frac{n!}{(n-a-b)!}\, f^a g^b \big(c + fx + gy\big)^{n-a-b} \qquad (a+b \le n), \tag{2.76}$$

and zero for $a + b > n$. Evaluating at the origin and applying (2.74),

$$p_{i,j}(t) = \frac{n!}{i!\, j!\, (n-i-j)!}\, f(t)^i g(t)^j c(t)^{\,n-i-j}, \qquad i + j \le n, \tag{2.77}$$

and $p_{i,j} = 0$ otherwise. This is the trinomial distribution, which is exactly what it should be: the n particles are independent, and at time t each is in one of three states, with probabilities c (neither inside nor outside — that is, dead), f (inside) and g (outside). Note in particular that the falling factorial $n!/(n-i-j)!$ in (2.76) is *not* n^{i+j}; the two agree only to leading order in n, and using the latter breaks normalisation.

Applied to the absorption models

For absorption only, (2.72), we have $c = 0$, $f = 1 - \mathrm{e}^{-\alpha t}$ and $g = \mathrm{e}^{-\alpha t}$. The factor c^{n-i-j} then vanishes unless $i + j = n$, and (2.77) gives

$$p_{i,j}(t) = \binom{n}{i} \big(1 - \mathrm{e}^{-\alpha t}\big)^i \big(\mathrm{e}^{-\alpha t}\big)^j, \qquad i + j = n,$$

so $X_t \sim \mathrm{Binomial}\big(n, 1 - \mathrm{e}^{-\alpha t}\big)$ and $Y_t = n - X_t$, as anticipated in section 2.A.1.

For absorption–death, (2.33), the three functions are

$$c(t) = 1 - \frac{\mu}{\mu - \alpha} \mathrm{e}^{-\alpha t} + \frac{\alpha}{\mu - \alpha} \mathrm{e}^{-\mu t}, \qquad f(t) = \frac{\alpha}{\mu - \alpha} \big(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\mu t}\big), \qquad g(t) = \mathrm{e}^{-\alpha t},$$

which are precisely $p_{0,0}$, $p_{1,0}$ and $p_{0,1}$ of (2.32), so that

$$p_{i,j}(t) = \frac{n!}{i!\, j!\, (n-i-j)!}\, f(t)^i g(t)^j c(t)^{\,n-i-j}.$$

Each particle independently ends up dead, absorbed, or still outside, with those three probabilities, and the joint law is trinomial. That the generating function factorises in this way is (2.46) read backwards.

The absorption–birth–death generating function of proposition 2.12 is not of the form (2.75), and there (2.74) must be applied to (2.63) directly, or the coefficients extracted numerically by Cauchy integrals on circles $|x| = |y| = \varrho < 1$.

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